Worktual Web Analytics

Database Tables — Retention & Purpose

Generated 2026-08-18 | ClickHouse: **web_tracking** | PostgreSQL: **web_analytics**

TTL note: ClickHouse deletes expired rows during background merges (not instantly). Query dimension tables with **FINAL**. Query gold tables with **countMerge()**, **uniqExactMerge()**, etc.

1. Retention summary (ClickHouse)

Table	Layer	Retention	TTL column	Queryable
raw_payloads	Bronze	90 days	received_at	No
events_staging	Pipeline / Staging	No TTL (recommended: 90 days)	received_at	No
dim_visitors	Dimension (Silver)	No TTL	—	Yes — use FINAL
dim_sessions	Dimension (Silver)	400 days	started_at	Yes — use FINAL
dim_domains	Dimension (Silver)	No TTL	—	No
admin_accounts	Legacy / Auth copy	No TTL	—	No
fact_page_views	Fact (Silver)	180 days	created_at	Yes
fact_interaction_events	Fact (Silver)	90 days	created_at	Yes
fact_form_events	Fact (Silver)	180 days	created_at	Yes
fact_conversion_events	Fact (Silver)	400 days	created_at	Yes
fact_custom_events	Fact (Silver)	180 days	created_at	Yes
fact_product_interactions	Fact (Silver)	180 days	created_at	Yes
fact_performance_events	Fact (Silver)	90 days	created_at	Yes
fact_error_events	Fact (Silver)	90 days	created_at	Yes
fact_cookie_operations	Fact (Silver)	60 days	created_at	No
fact_heap_snapshots	Fact (Silver)	30 days	created_at	No
session_replay_meta	Fact (Silver)	Not in repo DDL	—	Partial
session_replay_chunks	Fact (Silver)	Not in repo DDL	—	Partial
agg_web_daily	Aggregate (Gold)	3 years	event_date	Yes — use *Merge()
agg_page_daily	Aggregate (Gold)	2 years	event_date	Yes — use *Merge()
agg_product_daily	Aggregate (Gold)	2 years	event_date	Yes — use *Merge()

2. Full table catalog

Database	Table	Layer	Retention	TTL column	Queryable	Purpose
PostgreSQL	domains	Auth / Config	No automatic TTL	—	Dashboard admin	Registered websites. domain_id is the SDK tenant key; domain_name, site_type, status.
PostgreSQL	admin_accounts	Auth / Config	No automatic TTL	—	Login only	Dashboard admin users (email, password_hash) linked to one domain_id.
ClickHouse	raw_payloads	Bronze	90 days	received_at	No	Raw SDK JSON batches as received at ingest. Source for the entire MV pipeline.
ClickHouse	events_staging	Pipeline / Staging	No TTL (recommend 90 days)	received_at	No	Parsed events: one row per SDK event with props_json and profiles_json. Feeds all fact MVs.
ClickHouse	dim_visitors	Dimension (Silver)	No TTL	—	Yes — use FINAL	One row per visitor per domain. Lifetime total_sessions, total_pageviews, first/last seen, country.
ClickHouse	dim_sessions	Dimension (Silver)	400 days	started_at	Yes — use FINAL	One row per browsing session. UTM, referrer, device, geo, bounce, entry/exit pages.
ClickHouse	dim_domains	Dimension (Silver)	No TTL	—	No	ClickHouse copy of site metadata from ingest payloads. Auth source of truth is PostgreSQL domains.
ClickHouse	admin_accounts	Legacy / Auth copy	No TTL	—	No	Legacy ClickHouse copy of admin accounts. Not used — auth is PostgreSQL only.
ClickHouse	fact_page_views	Fact (Silver)	180 days	created_at	Yes	Each page load: page_path, title, time on page, scroll depth, referrer, UTM.
ClickHouse	fact_interaction_events	Fact (Silver)	90 days	created_at	Yes	Clicks, scroll, mouse move, rage clicks, dead clicks. Heatmap source (x/y, viewport, element).
ClickHouse	fact_form_events	Fact (Silver)	180 days	created_at	Yes	Form views, field focus, validation, submit, success, abandon.
ClickHouse	fact_conversion_events	Fact (Silver)	400 days	created_at	Yes	Goals, purchases, conversion value and type.
ClickHouse	fact_custom_events	Fact (Silver)	180 days	created_at	Yes	Custom SDK track() events and session replay payload references.
ClickHouse	fact_product_interactions	Fact (Silver)	180 days	created_at	Yes	E-commerce product views, clicks, price, category, SKU.
ClickHouse	fact_performance_events	Fact (Silver)	90 days	created_at	Yes	Web vitals: LCP, FCP, CLS, INP, navigation timing.
ClickHouse	fact_error_events	Fact (Silver)	90 days	created_at	Yes	JavaScript errors: message, stack, page_path, browser.
ClickHouse	fact_cookie_operations	Fact (Silver)	60 days	created_at	No	Cookie get/set/delete debug events. Reserved — not wired to dashboard.
ClickHouse	fact_heap_snapshots	Fact (Silver)	30 days	created_at	No	Browser memory heap snapshots. Debug only — not wired to dashboard.

Database	Table	Layer	Retention	TTL column	Queryable	Purpose
ClickHouse	session_replay_meta	Fact (Silver)	Not in repo DDL	—	Partial	Session replay session metadata (live DB only).
ClickHouse	session_replay_chunks	Fact (Silver)	Not in repo DDL	—	Partial	Session replay DOM/event chunks (live DB only).
ClickHouse	agg_web_daily	Aggregate (Gold)	3 years	event_date	Yes — use *Merge()	Pre-aggregated daily site KPIs: visitors, sessions, pageviews, clicks, bounce, conversions.
ClickHouse	agg_page_daily	Aggregate (Gold)	2 years	event_date	Yes — use *Merge()	Pre-aggregated daily metrics per page_path.
ClickHouse	agg_product_daily	Aggregate (Gold)	2 years	event_date	Yes — use *Merge()	Pre-aggregated daily metrics per product_id.

3. PostgreSQL (auth only)

Table	Purpose	Retention
domains	Registered websites. domain_id is the SDK tenant key; domain_name, site_type, status.	No automatic TTL
admin_accounts	Dashboard admin users (email, password_hash) linked to one domain_id.	No automatic TTL

4. Materialized views (pipeline — do not query directly)

View name	Flow	Purpose
mv_parse_events	raw_payloads → events_staging	Split JSON events[] into staging rows
mv_fact_page_views	events_staging → fact_page_views	Page view facts
mv_fact_interactions	events_staging → fact_interaction_events	Click/scroll/heatmap facts
mv_fact_form_events	events_staging → fact_form_events	Form funnel facts
mv_fact_performance	events_staging → fact_performance_events	Web vitals facts
mv_fact_errors	events_staging → fact_error_events	JS error facts
mv_fact_product	events_staging → fact_product_interactions	Product interaction facts
mv_fact_conversions	events_staging → fact_conversion_events	Conversion facts
mv_fact_custom	events_staging → fact_custom_events	Custom event facts
mv_dim_visitors	events_staging → dim_visitors	Visitor dimension upserts
mv_dim_sessions	raw_payloads → dim_sessions	Session dimension from batch profiles
mv_dim_domains	raw_payloads → dim_domains	Domain metadata sync
mv_agg_web_daily_pageviews	fact_page_views → agg_web_daily	Daily pageview aggregates
mv_agg_web_daily_interactions	fact_interaction_events → agg_web_daily	Daily click aggregates
mv_agg_web_daily_conversions	fact_conversion_events → agg_web_daily	Daily conversion aggregates
mv_agg_web_daily_sessions	dim_sessions → agg_web_daily	Daily session/bounce aggregates
mv_agg_page_daily	fact_page_views + interactions → agg_page_daily	Daily per-page aggregates
mv_agg_product_daily	fact_product_interactions → agg_product_daily	Daily per-product aggregates

5. Tables exposed to AI Chat & Dashboard API

SQL table	Display name
dim_visitors FINAL	Visitor Profiles
dim_sessions FINAL	Visitor Sessions
fact_page_views	Page View Events
fact_interaction_events	User Interaction Events
fact_conversion_events	Conversion Events
fact_product_interactions	Product Interaction Events
fact_custom_events	Custom Tracking Events
fact_form_events	Form Analytics Events
fact_performance_events	Web Performance Events
fact_error_events	JavaScript Error Events
agg_web_daily (*Merge)	Daily Site Metrics
agg_page_daily (*Merge)	Daily Page Metrics
agg_product_daily (*Merge)	Daily Product Metrics

Always filter: WHERE domain_id = <tenant> | **Session time:** dim_sessions.started_at | **Event time:** fact_*.created_at | **Daily trends:** agg_*.event_date